Version of October 28, 2026

MATH 6112. FALL 2026. ASSIGNMENT 3.

DUE: NOVEMBER 6, 11:59 PM

Instructions.

(i) You must return your homework typed on letter size paper. Text must be typed in 10pt font or larger.

(ii) The homework must be uploaded in the directory "Assignments" on Canvas by midnight (EDT) of the due day. Submission of a "pdf" file is required. Late submissions will be rejected.

(iii) Please, keep your arguments short. You do not need to rederive things we have done in class, and can just refer to the class notes. If in doubt, please ask.

(iv) Careful and precise mathematical writing is important. Arguments which are hard to follow will be marked down.

(v) The homework must reflect your own work. You are welcome to clarify the problems with one another and with me, but in the end you need to work on your own. Use of any source different from the class notes must be explicitly mentioned. This includes use of any AI tool, like `Chatgpt` or the like.

(vi) The homework is worth 105 points, but will be graded out of 100.

(vii) Note the **Due Date: November 6, midnight**.

For questions, W 3:30-5, or e-mail, or by appointment.

Homework 3 Problems.

(1) [*20 points.*] You are required to find counterexamples to the statements below, in order to dispel common misconceptions.

(a) *The product of two Hermitian matrices in* $\mathbb{C}^{n\times n}$, $n > 1$, *is Hermitian.*

(b) *Let* A *and* B *be Hermitian and suppose that* $0 \prec B \prec A$. *Then* $0 \prec B^2 \prec A^2$.

(c) *The symmetrized product of two positive definite matrices is positive definite.*

(d) *Let* $\Sigma(A)$ *denote the collection of singular values of a matrix* A, *and let* $B, C \in \mathbb{R}^{m\times n}, m \geq n$. *Then:* $\Sigma(CB^T) = \Sigma(B^TC) \cup \underbrace{\{0,\ldots,0\}}_{m-n}$.

(e) *Let* $A, B \in \mathbb{C}^{n\times n}$ *be Hermitian, and let their eigenvalues be ordered from smallest to largest:* $\lambda_1(A) \leq \lambda_2(A) \leq \cdots \leq \lambda_n(A)$, *and similarly for* B. *Assume that*

$$\lambda_i(A) \geq \lambda_i(B)\ , \quad \text{for all} \quad i = 1,\ldots,n\ .$$

Then: $A \succeq B$.

(2) [*10 points.*] In the notation of Problem (1)-(e), show that there always exists a unitary U so that $U^*AU \succeq B$.

(3) [*10 points.*] Let $A \in \mathbb{C}^{n\times n}$ with singular values $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$. Show that

$$|\operatorname{tr}(A)| \leq \sigma_1 + \cdots + \sigma_n\ .$$

(4) [*25 points.*] This is (most of) Exercise 5.2.29-(2), repeated here for convenience. Let $A, B \succeq 0$ and let at least one of them be positive definite. Consider the matrix

$$M = \begin{bmatrix} A & A^{1/2}B^{1/2} \\ B^{1/2}A^{1/2} & B \end{bmatrix}.$$

Show that $\det(M) = 0$ and that $M \succeq 0$. What is the dimension of the null space of M?

(5) [*25 points.*] This is Exercise 5.2.29-(3), repeated here for convenience. [Hint: Look at the Hint in the Lecture Notes, that will also introduce you to a useful technique.]
Let $A, B \succeq 0$. Show that

$$\det(A+B) \geq \det(A^{1/2}B^{1/2} + B^{1/2}A^{1/2})\ .$$

(6) [*15 points.*] Next, we introduce the Hadamard product of matrices (also called "pointwise product").

Definition [Hadamard Product] Let $A, B \in \mathbb{C}^{m\times n}$, then the *Hadamard product* of A and B, written $A \circ B$ is the matrix:

$$(A \circ B)_{ij} \;=\; a_{ij}b_{ij} \;,\; i = 1\ldots, m \;,\; j = 1, \ldots, n \;. \tag{0.1}$$

The Hadamard product is considerably easier to work with from the algebraic point of view than the standard matrix product, since it is associative and distributive (like standard matrix product), but also commutative (unlike standard matrix product). Obviously, it is also quite a bit less costly (in terms of arithmetic operation) to compute the Hadamard product rather than the standard product of matrices.

Some nice facts hold for Hadamard product, that do not hold for standard product of matrices; see Problem 1 of this Homework set.

Show the following result.

Let $A, B \in \mathbb{R}^{n\times n}$ be symmetric and $A \succeq 0$, $B \succeq 0$. Then $A \circ B \succeq 0$. Similarly, if $A \succ 0$ and $B \succ 0$ then $A \circ B \succ 0$. This result is known as **Schur's theorem**. [Hint: Let $D_x = \mathrm{diag}(x_1, \ldots, x_n)$ and write $x^T(A \circ B)x$ as the trace of the product of a matrix with its transpose.]